

Interactive Prototype Outcome Evaluation 2

Objective and validation metrics

For this evaluation I wanted to understand whether people could complete the Notion AR flow on a Meta Quest using only the right controller, and whether turning a 2D sketch into a 3D object still made sense once they were actually in the headset. I was also testing whether the controls felt instinctive, or whether they mainly made sense because I had chosen the buttons.

My testing focused on five areas:

Testing aim	Success metric
Complete the flow with the controller	Steps done without me explaining the controls. Ease rating from 1 to 5.
Find the most confusing steps	Steps marked Help or Fail, or named as the most confusing part.
Table is a clear workspace	Page placed on the table without help.
2D to 3D idea is understood	Reveal and lift done without help and the person can say what the step is for.
Concept seems useful	Usefulness rating from 1 to 5, plus comments.

I counted it as a success if participants could get through the main flow with limited help, place the page on the table without being told how, understand that the sketch was being lifted into a 3D object and give a usefulness rating that showed the idea was worth continuing.

I tested three participants for about five minutes each on a Meta Quest, using the right controller and reset with B between people. The sessions were informal tests with three friends who had already used the earlier desktop version. P3 was using VR for the first time. I did not explain the controls unless someone was stuck for about 10 seconds. Each step was marked Pass if they did it alone, Help if they finished after a hint, or Fail if they could not do it. After each test they rated ease and usefulness from 1 to 5.

Results

Three participants completed the same flow: open the page list, select the House Sketch, place it on the table, reveal the 2D sketch, lift it into a 3D house, and move the house around. The comments below are paraphrased from what they said. The raw notes are in Appendix A.

Step	P1	P2	P3
1. Open pages	Pass	Pass	Pass
2. Select House Sketch	Pass	Pass	Pass
3. Place on table	Pass	Pass	Pass
4. Reveal sketch	Pass	Pass	Pass
5. Lift into 3D	Pass	Pass	Help
6. Move house	Help	Help	Help
Ease (1–5)	4.5	4.3	3.5
Usefulness (1–5)	4.0	3.8	4.0

Overall, all three participants finished the flow and nobody failed a step. Steps 1 to 4 were a pass for everyone. P1 and P2 also lifted the house alone and found those steps easy and self-explanatory. P3 was slower on the lift because they were still finding the controller buttons, and needed a hint. The move step was a Help for all three. Each person kept tapping the grip instead of holding it and only moved the house after a hint. Ease averaged 4.1/5 (4.5, 4.3 and 3.5). Usefulness averaged 3.9/5 (4.0, 3.8 and 4.0). All three said the rest of the prototype was clear and that the controls should sit on the buttons you would expect.

Analysis and insights

The results show that the core concept still holds once it is on the Quest. People could open a page, put it on the table, reveal the sketch and for P1 and P2, lift it without me explaining the idea. Nobody was confused about what the prototype was for. The table still worked as a workspace. All three placed the page on it alone, and nobody named the table as confusing. The table gives the interface a clear physical place to work and that still held up in the headset.

A key pattern was that moving the house was harder than every earlier step. All three tapped the grip instead of holding it. This included P1 and P2, who found steps 1 to 5 easy and completed them alone. Every other step in the flow is a single press and nothing tells you that this one needs a hold. People did what the rest of the prototype had taught them to do. That makes it a design problem, not a skill problem.

Another important pattern was the button mapping. All three said the controls felt random and wanted actions on the buttons they already expect on a Quest: trigger or A to select, grip to grab. In this build, A

does two jobs, open and reveal and the trigger does three, select, place and lift. People were remembering a mapping rather than the controls explaining the task.

P3 shows a separate first time headset issue, but it does not explain the main problem. P3 was using VR for the first time and was slow at finding the controller buttons on the lift and the move. That is why ease dropped to 3.5, while usefulness stayed at 4.0. The controller is not visible in the headset, so knowing that a step uses A does not show where A is on the controller. One first time user is not enough to fully separate being new to VR from the design being unclear. The hold vs tap problem still hit P1 and P2, who had no trouble with the earlier steps so that finding is the stronger one.

Evaluation of aims

The aim of completing the flow with the controller was mostly validated. Average ease was 4.1/5 and all three finished. It was not a clean pass. Every participant needed a hint on the last item, P3 also needed one on the lift and all three found the mapping unnatural. The flow can be completed besides these problems but not with ease.

The aim of finding the most confusing steps was met. The confusing steps were hold versus tap when moving the house and for the new VR user, finding the controller buttons. Nothing else was named as confusing.

The table workspace aim was strongly validated. All three placed the page on the table without help and nobody named the table as a confusing step. Even though the table was understood in the next iteration the design can be upgraded to use the physical table in the user's environment.

The 2D to 3D aim was validated with some caution. All three revealed the sketch alone. P1 and P2 lifted it alone. P3's trouble on the lift was finding the button, not misunderstanding the idea. All three understood what the step was for. The caution is that everyone had already used the earlier version, so some of that understanding may have come from the previous test rather than from this prototype alone.

The usefulness aim was mostly validated. The average was 3.9/5 and the three scores sat close together, from 3.8 to 4.0. Nobody rejected the idea. The scores are positive rather than strong, which fits a test where the concept was clear and the controls were the thing that got in the way.

Concept iteration

Based on the evaluation, the next iteration will try to upgrade the table concept and keep the 2D to 3D lift and change the controls so the prototype matches what people already tried to do.

The next prototype should include:

1. **Optimised button mapping.** Trigger becomes select and confirm, so it covers select, place and lift. Grip becomes grab and move. A should have one job at most. B stays as the reset between participants. This comes directly from all three participants asking for the buttons they expect.
2. **A grab that matches the first attempt.** All three tapped instead of holding, so the next version should make that more clear with instructions.
3. **A controller hint for first time VR users.** Specifically direct users to what buttons to use instead of letting them guess. P3's lower ease came from not being able to see the controller, not from misunderstanding the task.
4. **Keep the step order, the reveal and the lift animation.** Those steps were completed alone by everyone, or for P3's lift, the problem was the button rather than the idea.

These changes are backed up with the testing results. I am keeping the parts that worked and changing the control scheme that did not. Notion AR stays a table workspace where a page's 2D sketch is lifted into a 3D object. The difference is that the controls follow normal Quest habits, the grab works the way people first try it, and a first time user can see which button to press.

Reflection and future testing

This evaluation helped me understand that moving the prototype onto the Quest answered the question the desktop test could not. People could actually do the flow in the headset and the confusion was about buttons, not about the concept. The table and the 2D to 3D idea held up. The control scheme did not. Testing on the headset, rather than only in the desktop simulator, was the right method for this question, because button finding and the grip hold only showed up once people were in the Quest.

I also learned that I had designed the buttons from a developer mindset. The mapping was easy to build and it made sense to me because I knew it. P1 and P2 completed every earlier step alone and still tapped instead of holding, so I cannot treat that as a one off mistake.

The method worked. Marking each step as Pass or Help, then collecting ease and usefulness, made the problem specific. I can point at one step rather than saying the prototype felt a bit confusing. What did not

work as well was the sample. I tested three friends who had already used the earlier version, not five new people. That likely made the flow feel easier and the ratings kinder. The move problem still showed up, which makes that finding stronger.

For the next round I want at least three new people who have not seen this prototype on the Quest, with the new button layout. I would watch whether they can complete the flow without a spoken walkthrough and whether a first time user can find the right button from the headset hint. I would also use a standard VR questionnaire so the ratings are easier to compare. That will tell me whether the concept is not only understandable, but usable without me standing next to the headset.

Appendix A: Raw participant notes

Participant 1

- Friend. Had used the earlier desktop version.
- Finished steps 1 to 5 alone and found them self-explanatory.
- On the move step, kept tapping the grip instead of holding it. Moved the house after a hint.
- Said the controls should be on more expected buttons (paraphrased).
- Ease of use: 4.5/5. Usefulness: 4.0/5.
- Most confusing: hold versus tap to move the house.

Participant 2

- Friend. Had used the earlier desktop version.
- Finished steps 1 to 5 alone and found them self-explanatory.
- On the move step, kept tapping the grip instead of holding it. Moved the house after a hint.
- Said the controls should be on more expected buttons (paraphrased).
- Ease of use: 4.3/5. Usefulness: 3.8/5.
- Most confusing: hold versus tap to move the house.

Participant 3

- Friend. First time using VR, but had used the earlier desktop version.
- Completed open, select, place and reveal alone.
- Lift was slowed by trouble finding the button, and was completed with guidance.
- Move: trouble finding the button, and kept tapping instead of holding. Completed with guidance.
- Otherwise found the flow self-explanatory.
- Said the controls should be on more expected buttons (paraphrased).
- Ease of use: 3.5/5. Usefulness: 4.0/5.
- Most confusing: finding the controller buttons, and hold versus tap to move.